

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_failure_replay.
- Best TSPLIB ATSP holdout gap: atsp_failure_replay.
- Best synthetic holdout gap: atsp_random_replay.

Run Metadata

- run_name: run_20260513_170341_g
- started_at_local: 2026-05-13 17:03:41
- finished_at_local: 2026-05-13 17:12:35
- duration_hhmm: 00:09
- duration_seconds: 533.825
- seed_offset: 6000
- replicate_label: g
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.174864	0.021916	0.09839	0.808852	0.78	0.032243
atsp_random_replay	random	score_only	False	0.189168	0.008454	0.098811	0.0	0.58	0.0
atsp_failure_replay	failure	score_only	False	0.127101	0.041579	0.08434	0.800347	0.78	0.030266
atsp_failure_replay_compression	failure	novelty_gate	True	0.193143	0.009004	0.101074	0.0	0.58	0.0

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.174864, synthetic 0.021916, combined 0.09839.

- Accepted-epoch count 2, mean accepted code novelty 0.808852, and final complexity 0.78.
- Adaptation efficiency 0.032243 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.174864 across 4 instances; family means: ft=0.287328, ftv=0.226286, p=0.001957, ry=0.183886.
- Panel synthetic_holdout mean gap 0.021916 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.069338, wind_clusters=0.018328.
- Worst recent training cases: [{"best_known_cost": 1286, "cost": 1551, "family": "ftv", "name": "ftv33", "optimality_gap": 0.206065}, {"best_known_cost": 1530, "cost": 1800, "family": "ftv", "name": "ftv38", "optimality_gap": 0.176471}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.189168, synthetic 0.008454, combined 0.098811.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.189168 across 4 instances; family means: ft=0.319768, ftv=0.33478, p=0.004982, ry=0.097143.
- Panel synthetic_holdout mean gap 0.008454 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.127101, synthetic 0.041579, combined 0.08434.
- Accepted-epoch count 3, mean accepted code novelty 0.800347, and final complexity 0.78.
- Adaptation efficiency 0.030266 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.127101 across 4 instances; family means: ft=0.223606, ftv=0.249225, p=0.004093, ry=0.03148.
- Panel synthetic_holdout mean gap 0.041579 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.080705, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1473, "cost": 1979, "family": "ftv", "name": "ftv35", "optimality_gap": 0.343517}, {"best_known_cost": 1286, "cost": 1649, "family": "ftv", "name": "ftv33", "optimality_gap": 0.282271}, {"best_known_cost": 1530, "cost": 1857, "family": "ftv", "name": "ftv38", "optimality_gap": 0.213725}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.193143, synthetic 0.009004, combined 0.101074.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.

- Panel heldout_tsplib mean gap 0.193143 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004093, ry=0.110456.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}, {"best_known_cost": 1286, "cost": 1478, "family": "ftv", "name": "ftv33", "optimality_gap": 0.1493}]

Judge Appendix

Summary of Replay Conditions on ATSP Benchmark Suite

Condition	Replay Mode	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Code Novelty Mean	Completeness	Notes on Transfer and Code Novelty
atsp_no_replay	none	0.174864	0.021916	0.09839	0.808852	0.78	Moderate transfer gap; high code novelty; no replay
atsp_random_replay	random	0.189168	**0.008454**	0.098811	0.0	0.58	Best synthetic holdout gap; transfer gap similar to no replay; no code novelty
atsp_failure_replay	failure	**0.127101**	0.041579	**0.08434**	0.800347	0.78	Best overall transfer (TSPLIB + transfer gap) despite higher synthetic gap; high code novelty maintained
atsp_failure_replay_compression	failure + compression	0.193143	0.009004	0.101074	0.0	0.58	Worst transfer gaps; compression pressure reduces code novelty and transfer

Conservative Interpretation

- ****Transfer Performance (Main Evidence):****
- ***Held-out TSPLIB Gap*** (primary transfer metric): The ****atsp_failure_replay**** condition achieves the ****lowest mean TSPLIB gap (0.1271)****, indicating superior transfer to held-out ATSP instances.
- ***Synthetic Holdout Gap***: ****atsp_random_replay**** achieves the best synthetic evaluation gap (0.00845), showing strong synthetic holdout optimality but its TSPLIB gap is higher (0.189168).
- ***Mean Transfer Gap*** confirms ****atsp_failure_replay**** is best overall (0.08434), reinforcing confidence in that replay mode for enabling transfer.
- ****Code Novelty vs Transfer:****
- The ****failure replay without compression**** maintains high code novelty (~0.80), similar to no replay (~0.81), yet achieves improved transfer metrics (TSPLIB gap reduced from 0.1749 to 0.1271, a significant gain).
- ****Random replay**** and ****failure replay with compression**** have zero code novelty but show worse or equivalent transfer gaps compared to failure replay without compression.
- This indicates that ****code novelty decreases in random replay and compression-aware replay conditions, while transfer does not improve or even deteriorates in those cases****.

- Thus, **improved transfer with failure replay occurs alongside maintained code novelty, whereas lowering code novelty (random, compressed replay) does not yield transfer benefits.**
- **Replay Mode Impact:**
- **No Replay (none):** Acceptable transfer performance with high novelty.
- **Random Replay:** Best synthetic gap but worse transfer gaps; no code novelty; reduced complexity.
- **Failure Replay (best_holdout & best_transfer):** Best held-out and transfer performance; moderate synthetic gap; high code novelty; complex behavioral profile.
- **Failure Replay + Compression:** Worst transfer and TSPLIB performance; minimal novelty; compression pressure likely harms transfer.
- **Complexity and Archive Sizes:**
- Failure replay conditions show higher complexity (0.78) and larger elite archive (3) compared to random and compressed conditions (0.58 complexity, archive size 1).
- Compression pressure in failure replay compresses behavior and code novelty, hurting transfer.

Key Takeaways

- **Failure Replay (without compression) maximizes transfer performance to held-out TSPLIB ATSP problems, yielding the lowest optimality gaps in transfer benchmarks.**
- **This improvement in transfer is accompanied by sustained high code novelty and behavioral complexity, not reduced code novelty.**
- **Random replay improves synthetic holdout gap but fails to improve actual transfer (held-out TSPLIB), and shows zero code novelty.**
- **Failure replay with compression reduces code novelty and transfer quality, indicating compression pressure negatively impacts transfer despite replay.**
- **No replay and failure replay both show high code novelty, but failure replay improves transfer metrics clearly.**

Recommendation

- Prioritize **failure replay without compression** for optimal transfer performance in replay-aware ATSP benchmarks.
- Avoid compression-induced novelty gating during failure replay as it detrimentally affects transfer learning.
- Recognize that algorithmic invention (as measured by transfer and performance gains) correlates here with maintained or elevated code novelty rather than lexical novelty reduction.
- Interpret synthetic holdout improvements cautiously and prioritize TSPLIB held-out gaps as main evidence for transfer success.

Let me know if you need numerical drilling into individual instance results or deeper analysis of complexity/adaptation.